

# Adding and Subtracting Fractions with Like Denominators

## Answer Key

### Grades 4–5

#### Quick Answers

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1.  $\frac{4}{5}$   
 2.  $\frac{7}{8}$   
 3.  $\frac{1}{3}$   
 4.  $\frac{2}{3}$

5.  $\frac{9}{1000}$   
 6.  $\frac{1}{1000}$   
 7.  $\frac{1}{1000}$   
 8.  $\frac{5}{6}$

9.  $\frac{7}{8}$   
 10.  $\frac{5}{7}$

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#### Curriculum

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**Level:** Grades 4–5

4.NF.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

*Difficulty 1–4 of 5 across 10 tagged checks.*

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**1. Fifths bar:**  $\frac{1}{5} + \frac{3}{5}$ .

**Solution:** One fifth is 1 shaded part; three fifths is 3 more parts of the same size. On the bar that is  $1 + 3 = 4$  shaded parts out of 5, so only the count of parts changes — the size of a part is still one fifth.

$$\frac{1}{5} + \frac{3}{5} = \frac{1+3}{5} = \boxed{\frac{4}{5}}$$

Already in simplest form (4 and 5 share no factor but 1).

**2. Eighths bar:**  $\frac{2}{8} + \frac{5}{8}$ .

**Solution:** Count eighths: 2 shaded, then 5 more shaded, giving  $2 + 5 = 7$  eighths. The denominator stays 8 because the bar is still cut into 8 equal parts.

$$\frac{2}{8} + \frac{5}{8} = \frac{2+5}{8} = \boxed{\frac{7}{8}}$$

The bar is not quite full: 7 of the 8 parts are shaded.

**3. Ninths bar:**  $\frac{7}{9} - \frac{4}{9}$ .

**Solution:** Start with 7 shaded ninths and cross out 4 of them.  $7 - 4 = 3$  ninths are still shaded.

$$\frac{7}{9} - \frac{4}{9} = \frac{7-4}{9} = \frac{3}{9}$$

Now simplify: 3 and 9 are both divisible by 3, and  $3 \div 3 = 1$ ,  $9 \div 3 = 3$ .

$$\boxed{\frac{1}{3}}$$

On the bar, the 3 shaded ninths line up exactly with one third of the bar, which is why the two names describe the same amount.

**4. Sixths bar:**  $\frac{5}{6} - \frac{1}{6}$ .

**Solution:**  $5 - 1 = 4$  sixths remain shaded.

$$\frac{5}{6} - \frac{1}{6} = \frac{4}{6}$$

Both 4 and 6 divide by 2, so  $\frac{4}{6} = \boxed{\frac{2}{3}}$ .

**5. Two eighths bars:**  $\frac{5}{8} + \frac{4}{8}$ .

**Solution:** Counting eighths across both bars:  $5 + 4 = 9$  eighths.

$$\frac{5}{8} + \frac{4}{8} = \boxed{\frac{9}{8}}$$

Nine eighths is more than one whole. Eight of the nine eighths fill one whole bar and one eighth is left over, so the same amount is 1 whole and 1 eighth.

*Watch for:* an answer of  $\frac{9}{16}$  — that comes from adding the denominators, which would mean the pieces got smaller when nothing was cut.

**6. Missing numerator:**  $\frac{2}{9} + \underline{\quad} = \frac{7}{9}$ .

**Solution:** The pieces are all ninths, so the question is really “2 ninths plus how many ninths makes 7 ninths?” Count up on the bar from 2 to 7: that is  $7 - 2 = 5$  more ninths.

$$\frac{2}{9} + \frac{5}{9} = \frac{7}{9} \quad \checkmark \quad \boxed{\frac{5}{9}}$$

**7. Chocolate bar story:**  $\frac{11}{12} - \frac{5}{12}$ .

**Solution:** “Eats” means take away, so subtract. The bar is cut into 12 equal pieces, so both fractions count twelfths and the denominator does not change.

$$\frac{11}{12} - \frac{5}{12} = \frac{11-5}{12} = \frac{6}{12}$$

Simplify by dividing both parts by 6:  $\boxed{\frac{1}{2}}$  of the bar is left.

*Watch for:*  $\frac{16}{12}$  — that is what adding gives, and an answer larger than what Lena started with cannot be what is left.

**8. Three shaders on one sixths bar:**  $\frac{1}{6} + \frac{2}{6} + \frac{2}{6}$ .

**Solution:** All three friends shade sixths of the same bar, so add the numerators only:  $1 + 2 + 2 = 5$ .

$$\frac{1}{6} + \frac{2}{6} + \frac{2}{6} = \boxed{\frac{5}{6}}$$

The bar is *not* full: 5 of its 6 parts are shaded, so one sixth is still blank.

**9. Work backwards:**  $\_ - \frac{3}{8} = \frac{4}{8}$ .

**Solution:** The amount before pouring is the amount poured out plus the amount still there — undo the subtraction by adding.

$$\frac{3}{8} + \frac{4}{8} = \frac{7}{8}$$

Check by subtracting:  $\frac{7}{8} - \frac{3}{8} = \frac{4}{8}$  ✓. The jug held  $\boxed{\frac{7}{8}}$  of a jug before the pouring.

**10. Find and fix:** Jae wrote  $\frac{3}{7} + \frac{2}{7} = \frac{5}{14}$ .

**Solution:** Jae added the denominators as well as the numerators. That cannot be right:  $\frac{5}{14}$  is smaller than  $\frac{3}{7}$ , and adding a positive amount can never make the total smaller. On a sevenths bar, shading 3 parts and then 2 more leaves 5 shaded parts, and the parts are still sevenths — nothing re-cut the bar into fourteenths.

$$\frac{3}{7} + \frac{2}{7} = \frac{3+2}{7} = \boxed{\frac{5}{7}}$$